PART 1

Regression Analysis

Predicting Athlete Performance Score

Dataset: Kaggle — Athlete Performance Evaluation | 1,000 records | 12 features

Majd | DS301 Capstone

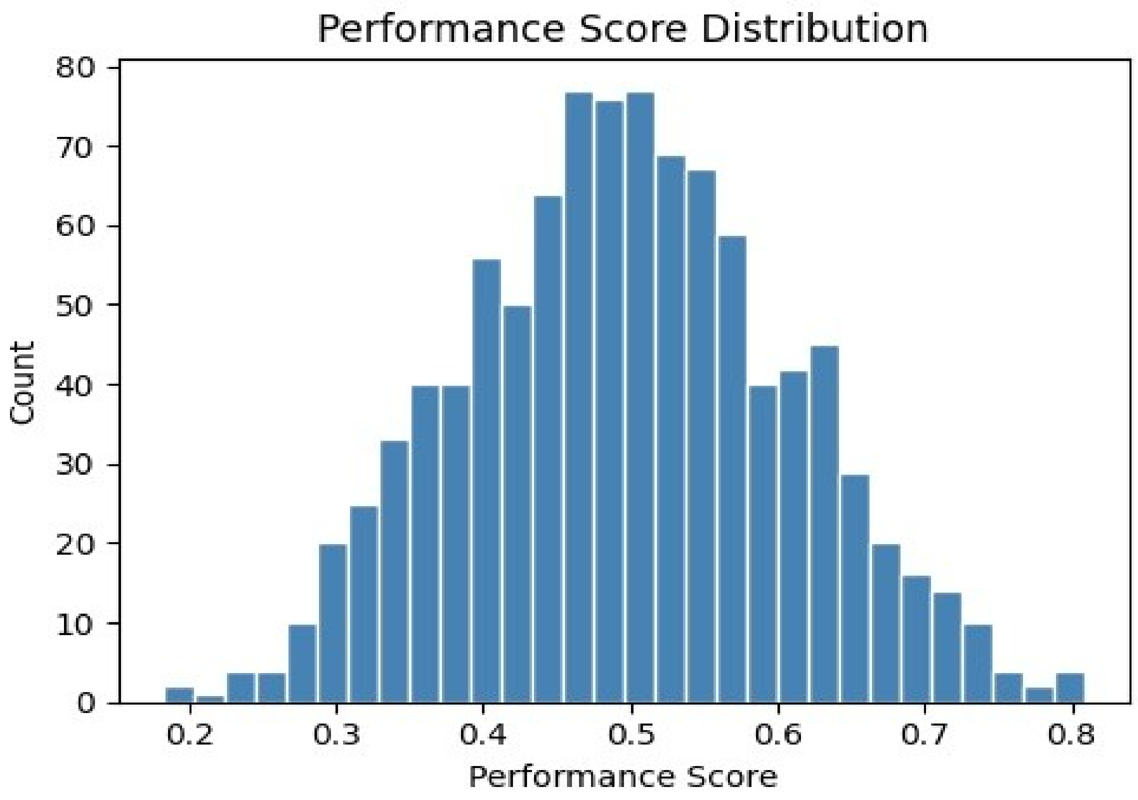
Dataset Overview & Target Distribution

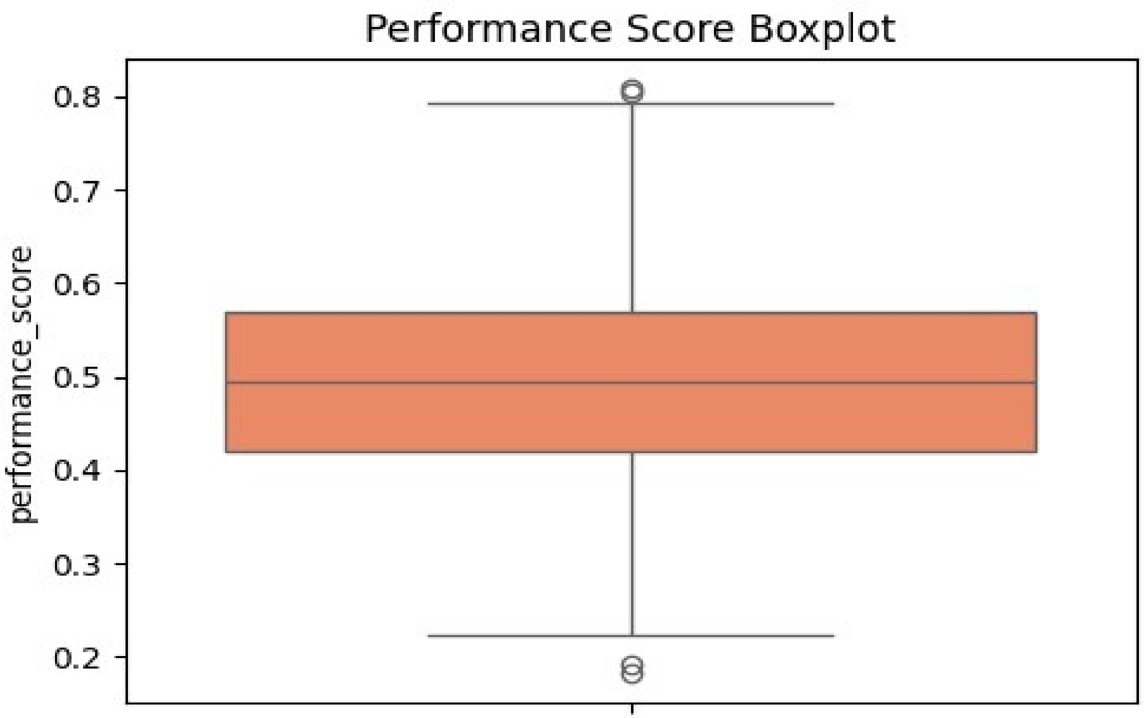
1,000
Records

12
Features

0
Missing Values

80/20
Train/Test Split





7 Required Regression Models

1 Linear Regression

Baseline — no regularization

2 Lasso (L1)

alpha=0.01, feature selection

3 Ridge (L2)

alpha=1.0, recommended model

4 SVR

RBF kernel, StandardScaler Pipeline

5 Decision Tree

random_state=42

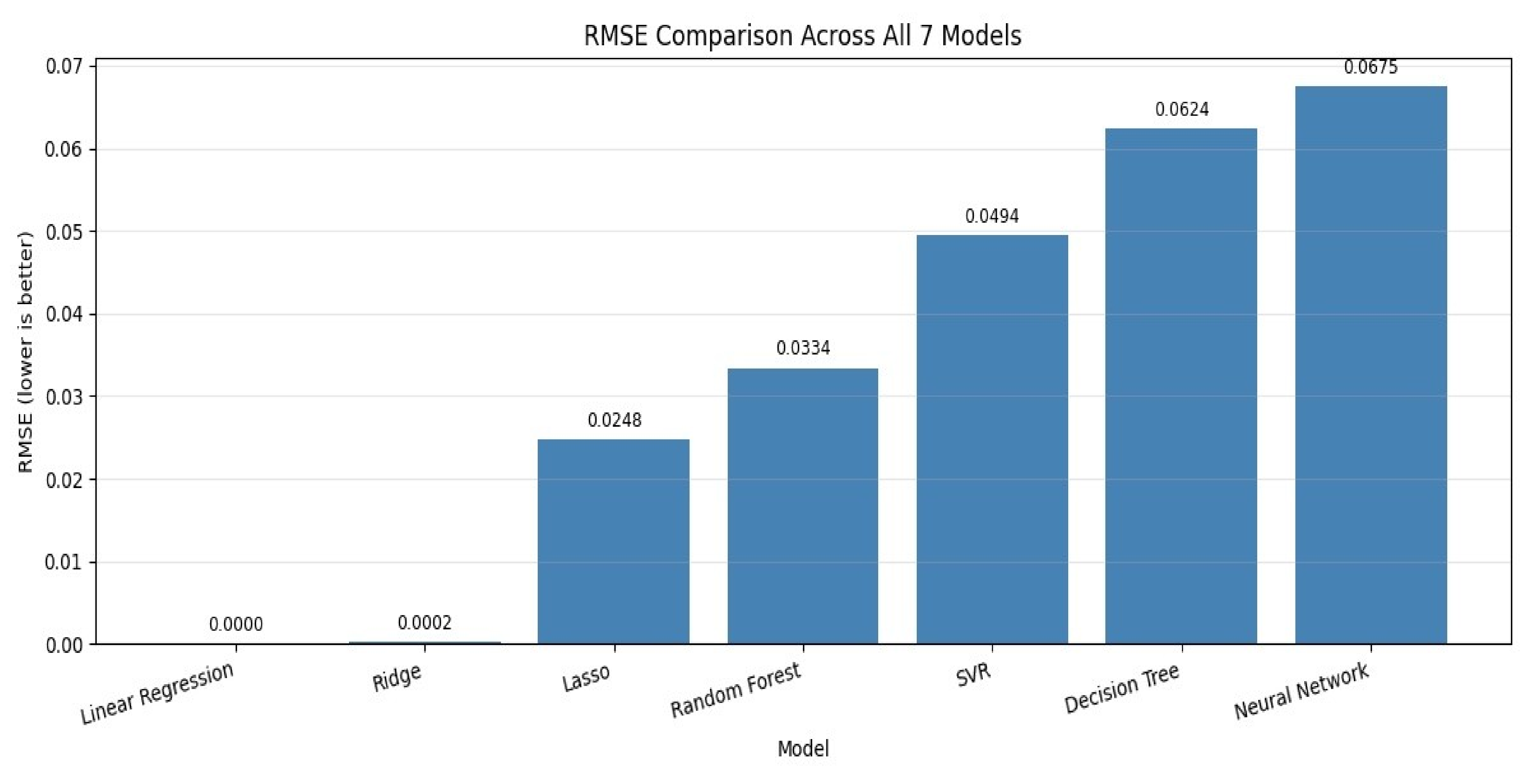
6 Random Forest

100 trees, feature importances

7 Neural Network

Keras: Dense(64)→Dense(32)→Dense(1) EarlyStopping, loss=mse

RMSE Comparison — All 7 Models



Model Performance Results

Model	RMSE	MAE	R^2	Train Time (s)
Linear Regression	0.0000	0.0000	1.0000	0.0080
Ridge Regression ⭐	0.0002	0.0001	1.0000	0.0059
Lasso Regression	0.0248	0.0202	0.9507	0.0066
Random Forest	0.0334	0.0277	0.9104	0.4666
SVR	0.0494	0.0371	0.8047	0.0077
Decision Tree	0.0624	0.0502	0.6877	0.0095
Neural Network (Keras)	0.0675	0.0527	0.6349	0.9341
Random Forest (Tuned)	0.0325	0.0267	0.9155	—

⭐ *Best Model: Ridge Regression — near-perfect accuracy, fast training, robust L2 regularization*

Hyperparameter Tuning — Random Forest

GridSearchCV Setup

Method: GridSearchCV (5-fold CV)

Model: Random Forest Regressor

Fit on training data ONLY

Parameter Grid:

n_estimators = [50, 100, 200]

max_depth = [None, 5, 10]

min_samples_split = [2, 5]

Total combinations: 18

Best Parameters Found

n_estimators	**200**
max_depth	**10**
min_samples_split	**2**

Results

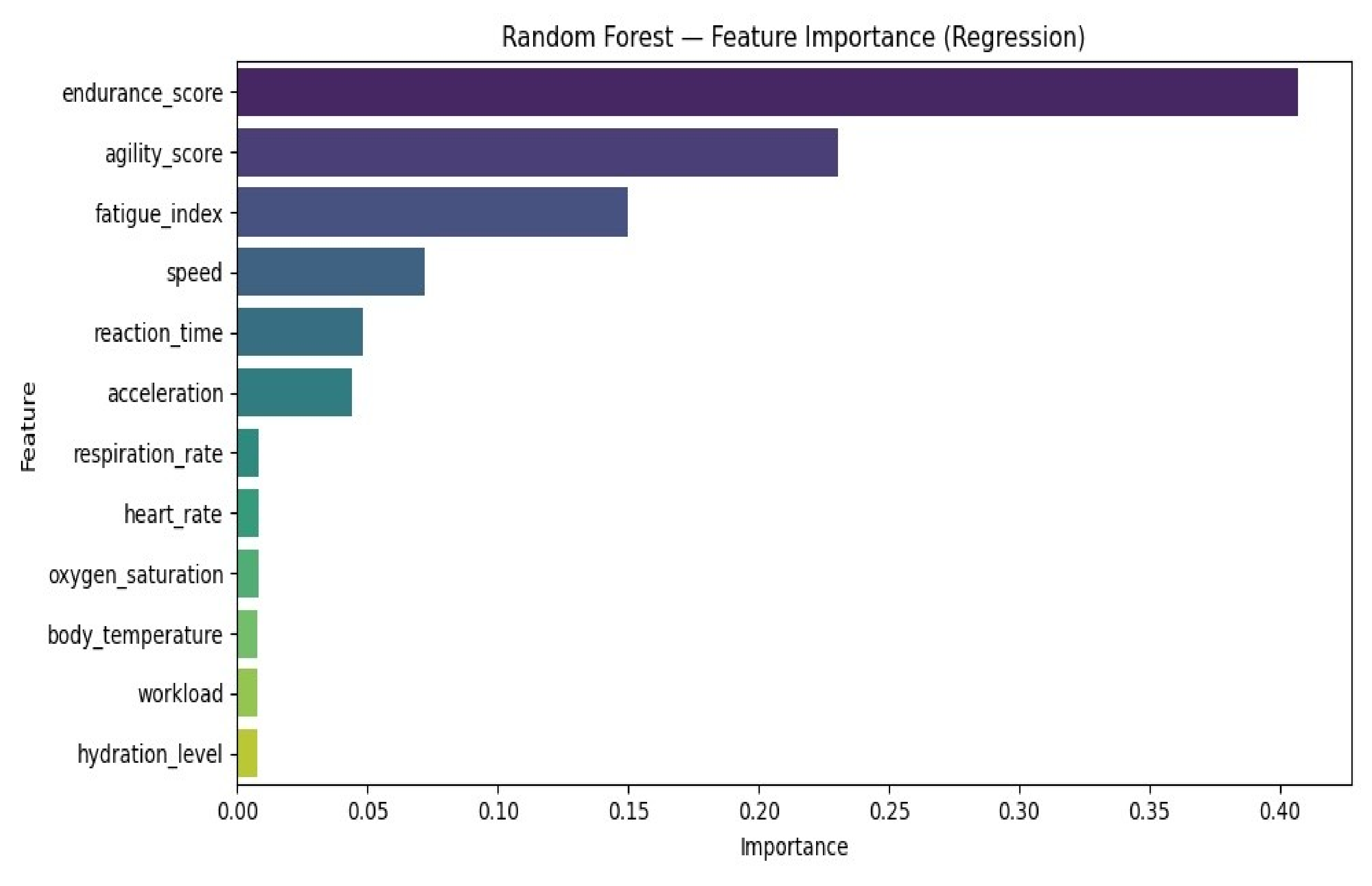
Default RF

RMSE: 0.0334 R^2: 0.9104

Tuned RF ⭐

RMSE: 0.0325 R^2: 0.9155

Feature Importance — Random Forest



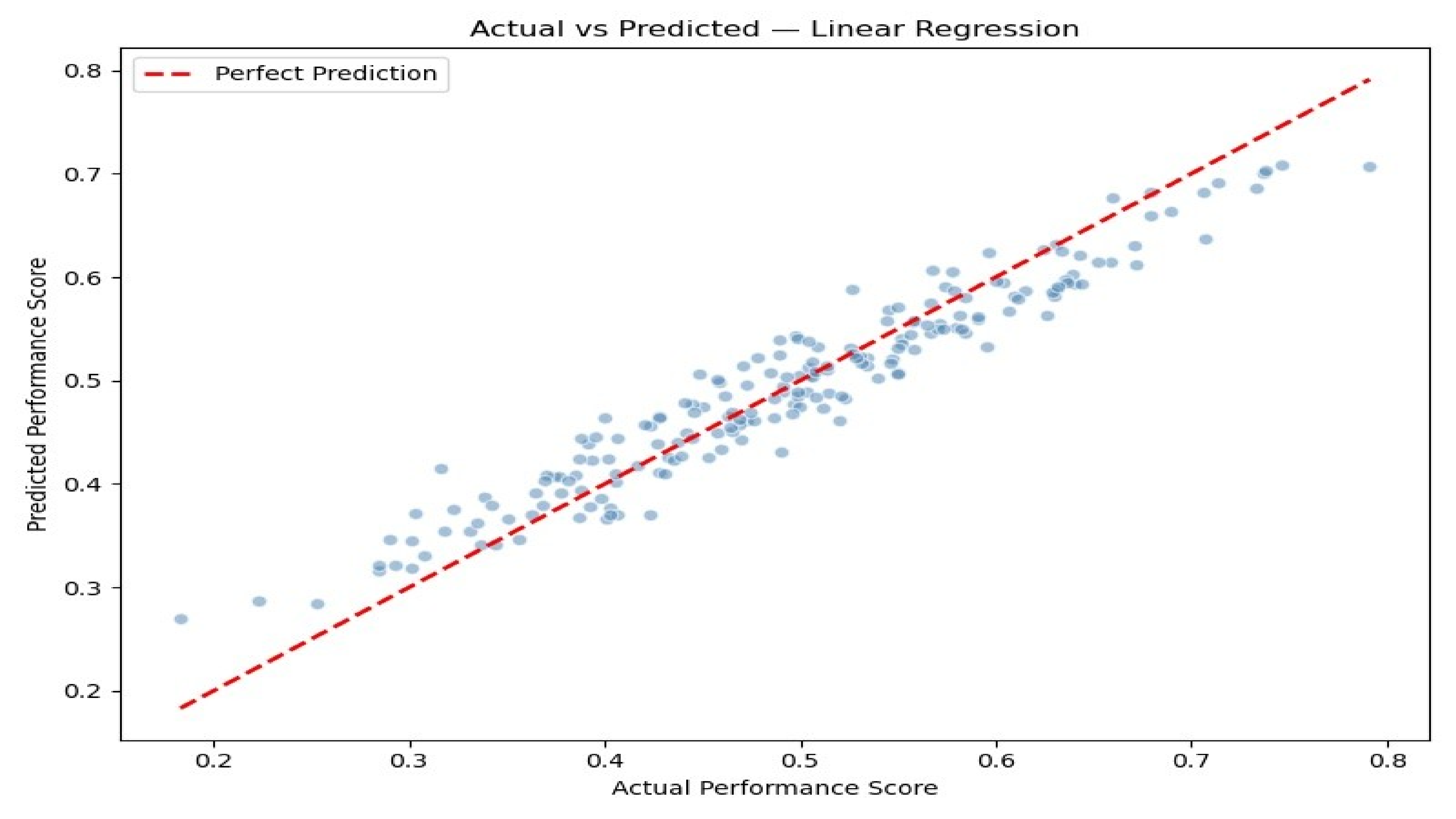
Key Insight

endurance_score	40.7%
agility_score	23.0%
fatigue_index	15.0%

Top 3 features account for ~79% of predictive power.

Physiological features (heart rate, body temp, oxygen) have minimal impact.

Actual vs. Predicted — Best Model (Ridge Regression)



Note: Plot shows Linear Regression predictions which closely match the diagonal (perfect prediction line), confirming near-perfect R^2=1.0000 performance.

Conclusion

7 Models Trained

Linear, Lasso, Ridge, SVR,
Decision Tree, Random Forest,
Neural Network (Keras)

Best: Ridge Regression

RMSE=0.0002, R^2=1.0000
Fastest robust model (0.006s)

Tuning Improved RF

GridSearchCV: RMSE
0.0334 → 0.0325

Key Feature

Endurance score drives
40.7% of all predictions